

not read the strength accurately and can be misleading, because it registers the strength of the strongest signal, which may simply be the closest computer. The IEEE 802.11 specification includes provisions designed to minimise collisions between two independent transmissions. Two mobile units may both be in range of a common access point, but not in range of each other. Therefore, the 802.11 standard has two basic modes of operation: ad hoc mode enables peer-to-peer transmission between mobile units, and using the infrastructure mode, mobile units communicate through an access point that serves as a bridge to a wired network infrastructure. Since wireless communications use a more open medium for communication in comparison to wired LANs, the 802.11 designers also included shared-key encryption mechanisms. Some examples are the Wired Equivalent Privacy (WEP) and the Wi-Fi Protected Access (WPA, WPA2) which are used to ensure security.

Bridge: a bridge can be used to connect different networks.

Roaming is not limited to cellular networks. Even wireless LANs encounter roaming

A wireless Ethernet bridge allows the connection of devices on a wired Ethernet network to a wireless network. The bridge acts as the connection point to the Wireless LAN. This way, any device within range of another device with an internet

connection can access the internet.

Wireless distribution system: a wireless distribution system allows the wireless interconnection of access points in an IEEE 802.11 network. If a proper distribution network is set up, then a wireless network can be expanded using multiple access points without the need for a wired backbone system to link them. The advantage of a distribution system is that it preserves the MAC addresses of clients.

Roaming: in a wireless local area network, there are two types of roaming. Internal roaming occurs when the mobile station moves from one access point to another within, say, a home network. If the network signal is too weak and connectivity is not maintained during the switch from one access point to another, the data transfer will be interrupted. Hence, there are software solutions to maintain connections.